Results: completeness of relational operations in LLMs

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Setup. 500 questions asking for *every* item that satisfies a condition, over synthetic records whose correct answers come from executing SQL. They cover enumeration and six SQL-like operations, in three settings: records in the prompt (400 questions), memory only (60), and retrieval (40). Five models were tested (DeepSeek V4 Pro and Flash, Claude Haiku 5.5, MiMo-V2.6 Pro and Flash), with three runs:

- **E1:** all 500 questions with a direct prompt.
- **E2:** two alternative prompts on a 100-question subset.
- **E3:** the same subset run a second time.

Exact-set accuracy is the share of answers that are exactly the correct set. Errors are counted as *missing*, *near-miss* (a garbled copy of a correct item), *wrong* (in the data but not matching), *made-up* (not in the data) or *duplicate*. Brackets are 95% intervals. Unless stated otherwise, results are for the 365 in-context questions with list answers, in E1.

1. Overall completeness

Recall is close to 1 for every model, but only 81–98% of answers are exactly the correct set: incomplete answers usually miss only a few items (median 1 per incomplete answer). The best model, DeepSeek V4 Flash, is statistically tied with DeepSeek V4 Pro; the other three are significantly lower, by 7 to 17 percentage points.

Model	Exact-set acc. (%)	Δ vs. best (pp)	Mean recall	Mean prec.	Omit. (%)	Comm. (%)	Invalid
DeepSeek V4 Pro	98 [96, 99]	−0.3 [-2.5, +1.6]	0.996 [0.991, 1.000]	1.000	2	0	1
DeepSeek V4 Flash	98 [96, 99]	(reference)	0.996 [0.989, 1.000]	0.999	1	1	1
Claude Haiku 5.5	91 [88, 93]	−7.1 [-10.1, -4.1]	0.995 [0.993, 0.997]	0.999	9	1	0
MiMo-V2.6 Pro	90 [86, 93]	−8.2 [-11.8, -4.9]	0.987 [0.978, 0.994]	0.997	7	4	2
MiMo-V2.6 Flash	81 [77, 85]	−16.7 [-20.8, -12.3]	0.966 [0.952, 0.979]	0.976	15	10	2

Table 1: Δ: paired difference in exact-set accuracy from the best model. Mean recall counts invalid answers as 0; precision is over valid answers. *Omit./Comm.*: share of answers that leave out a correct item / add an incorrect or duplicate one. *Invalid*: truncated, unparseable or refused.

2. Error types

Four of the five models almost only *omit* items: together they returned 25 wrong items and 0 made-up ones. MiMo-V2.6 Flash also *invents* items on long lists: 1,032 made-up items in total, 991 of them on a single 100-item question. Box 1 shows one example of each.

Model	Missing	Near-miss	Wrong	Made-up	Duplicate	Invalid answers
DeepSeek V4 Pro	6	0	1	0	0	1
DeepSeek V4 Flash	4	1	3	0	0	1
Claude Haiku 5.5	75	0	6	0	0	0
MiMo-V2.6 Pro	49	0	15	0	0	2
MiMo-V2.6 Flash	342	7	311	1,032	316	2

Table 2: Error counts by type, on valid answers; the last column counts the invalid answers excluded.

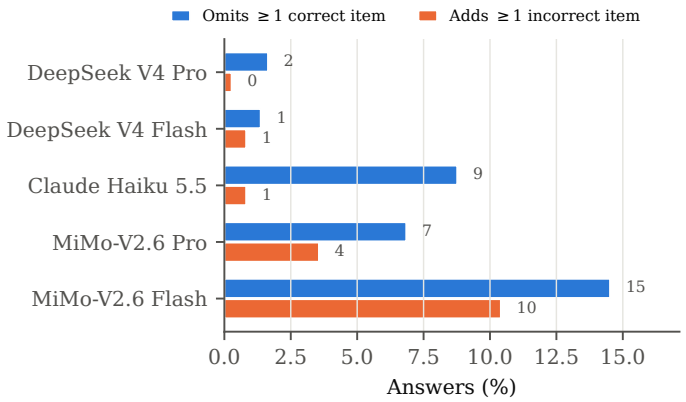


Figure 1: Share of answers that omit at least one correct item, and that add at least one incorrect item.

Box 1. Two failures

Omission (DeepSeek V4 Pro, `enum-ctx-029`). Prompt: 500 employee records as CSV (≈8k tokens), then *"List the names of all employees whose department is Support."* Gold answer: 50 names. The model returned 49 names, all correct, and omitted Kenneth Grimes (row 321 of 500). In three re-runs the answer was incomplete every time, omitting 2–7 names, a different set each time (Jose Greene in all three), and two of the three re-runs also added incorrect names.

Fabrication (MiMo-V2.6 Flash, `enum-ctx-053`). Prompt: 1,000 book records; *"List the titles of all books whose genre is Romance."* Gold answer: 100 titles. The model returned 1,125 distinct titles (plus 245 repeats): 66 correct, 68 real but not matching, and 991 that do not exist in the data, for example "Blooming Season", "Cold Garden", "Crimson Orchad", "Ember of Empty Phevi".

3. Answer size and table size

Completeness falls as the answer set grows, and separately as the table grows. With up to 10 correct items, every model is exact on at least 93% of questions. With 50 or more, Claude Haiku 5.5 drops to 53% and MiMo-V2.6 Flash to 34%. Pooled across models, each doubling of the answer set multiplies the odds of a complete answer by 0.50 (0.36–0.60), and each doubling of the table by 0.39 (0.22–0.56); both estimates are adjusted for each other and for input format. The DeepSeek models fail too rarely (4 and 2 of 100) for per-model estimates.

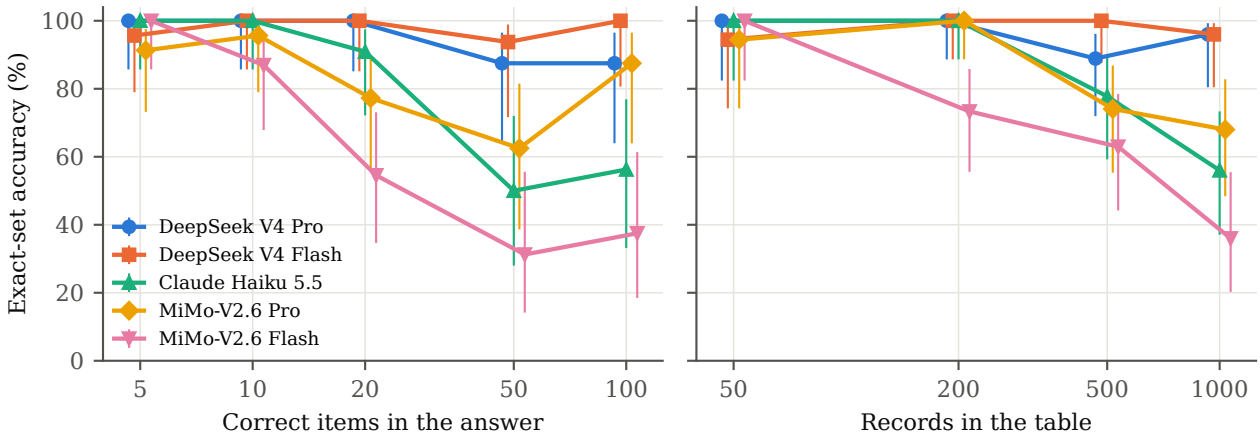


Figure 2: List-all-matches questions in context: exact-set accuracy by number of correct items (left) and table size (right).

Model	≤10 items (%)	≥50 items (%)	Failures	OR: answer size [95% CI]	OR: table size [95% CI]
DeepSeek V4 Pro	100	88	4	n.e.	n.e.
DeepSeek V4 Flash	98	97	2	n.e.	n.e.
Claude Haiku 5.5	100	53	17	0.17 [0.02, 0.28]	0.07 [0.01, 0.17]
MiMo-V2.6 Pro	93	75	16	0.79 [0.44, 1.20]	0.44 [0.11, 0.76]
MiMo-V2.6 Flash	93	34	34	0.33 [0.15, 0.44]	0.29 [0.08, 0.45]
All models (pooled)	97	69	73	0.50 [0.36, 0.60]	0.39 [0.22, 0.56]

Table 3: Odds ratio of a complete answer per doubling (below 1 = worse), from one logistic regression with answer size, table size and input format. n.e.: not estimable (fewer than 5 failures).

4. By operation

Every operation except set operations produced incomplete answers from at least three models. Enumeration, which has the largest answer sets, is hardest.

Operation	Qs	DeepSeek V4 Pro	DeepSeek V4 Flash	Claude Haiku 5.5	MiMo-V2.6 Pro	MiMo-V2.6 Flash
Enumeration	100	96	98	83	84	66
Selection	60	97	98	95	93	88
Projection	40	100	95	90	90	82
JOIN	60	100	98	93	88	82
Aggregation	15	93	93	93	87	67
Nested query	40	98	100	90	92	90
Set operation	50	100	100	100	98	100

Table 4: Exact-set accuracy (%) by operation. Aggregation here is GROUP BY; single-number aggregates are in section 8.

5. Input format

Format matters for some models only. For example, MiMo-V2.6 Flash is exact on 73% of questions given as English sentences against 86% as JSON Lines. The DeepSeek models are unaffected.

Input format	Qs	DeepSeek V4 Pro	DeepSeek V4 Flash	Claude Haiku 5.5	MiMo-V2.6 Pro	MiMo-V2.6 Flash
Markdown	105	98	98	88	90	87
CSV	97	96	97	93	93	78
JSON Lines	81	99	100	96	83	86
English sentences	82	99	98	88	94	73

Table 5: Exact-set accuracy (%) by how the records were shown. Formats rotate across questions.

6. Number of conditions

More WHERE conditions did not make answers less complete. But questions with more conditions match fewer rows (median 13, 6.5 and 3.5 items), so this can't yet be separated from answer size.

Conditions	Qs	Median answer size	DeepSeek V4 Pro	DeepSeek V4 Flash	Claude Haiku 5.5	MiMo-V2.6 Pro	MiMo-V2.6 Flash
1	28	13	96	100	93	89	86
2	20	6.5	95	95	95	95	90
3	12	3.5	100	100	100	100	92

Table 6: Exact-set accuracy (%) on selection questions by number of WHERE conditions.

7. Memory-only and retrieval settings

From memory, the DeepSeek models were always exact. The other three missed members of short fixed lists, e.g. Claude Haiku 5.5 named Kentucky but not New Jersey as a state ending in "y". In the retrieval setting, every model returned every correct record it was given, so recall (0.869 for all models) only reflects what the retriever found.

Model	Memory: exact (%)	Memory: recall	Retrieval: recall	Retrieval: recall over retrieved
DeepSeek V4 Pro	100	1.000	0.869	1.000
DeepSeek V4 Flash	100	1.000	0.869	1.000
Claude Haiku 5.5	87	0.952	0.869	1.000
MiMo-V2.6 Pro	83	0.955	0.869	1.000
MiMo-V2.6 Flash	83	0.958	0.869	1.000

Table 7: Memory-only and retrieval questions. "Recall over retrieved" counts only correct items the retriever returned.

8. Counts and totals

On 35 COUNT, SUM, MIN, MAX and AVG questions, the DeepSeek models were always exact. Of the other models' 11 wrong answers, 9 were too low, which is consistent with items being dropped before counting.

Model	Qs	Exact (%)	Under-counts	Over-counts
DeepSeek V4 Pro	35	100	0	0
DeepSeek V4 Flash	35	100	0	0
Claude Haiku 5.5	35	94	2	0
MiMo-V2.6 Pro	35	91	3	0
MiMo-V2.6 Flash	35	83	4	2

Table 8: Single-number aggregation questions.

9. Consistency across runs

A second run of the same 100 prompts gave the same verdict on 87–100% of questions. Some failures recur every time: Claude Haiku 5.5 left Maine out of "states containing 'in' " in all four runs, and DeepSeek V4 Pro was incomplete on the same 50-item list in all four. Others never recur. So completeness is best reported as a rate over many questions. The failure cases in Table 10 were chosen by hand.

Model	Qs	Same verdict (%)	Questions that flipped
DeepSeek V4 Pro	87	98 [92, 99]	2
DeepSeek V4 Flash	93	100 [96, 100]	0
Claude Haiku 5.5	93	94 [87, 97]	6
MiMo-V2.6 Pro	93	91 [84, 96]	8
MiMo-V2.6 Flash	93	87 [79, 92]	12

Table 9: Agreement between two independent runs (E1 vs. E3).

Model	Question	Answers	Records	Main-run outcome	Reproduced
Claude Haiku 5.5	`enum-par-013`	2	50	recall 0.50, 1 missing	2/3
Claude Haiku 5.5	`enum-par-021`	11	50	recall 0.73, 3 missing	3/3
Claude Haiku 5.5	`sel-029`	13	400	recall 0.85, 2 missing	0/3
Claude Haiku 5.5	`join-004`	55	100	recall 0.93, 4 missing	0/3
Claude Haiku 5.5	`nest-022`	13	200	recall 0.92, 1 missing	2/3
Claude Haiku 5.5	`proj-031`	37	300	recall 0.89, 4 missing	0/3
Claude Haiku 5.5	`agg-015`	12	400	answered 11 (gold 12)	0/3
Claude Haiku 5.5	`enum-ctx-029`	50	500	recall 0.96, 2 missing	0/3
DeepSeek V4 Pro	`enum-ctx-029`	50	500	recall 0.98, 1 missing	3/3
MiMo-V2.6 Flash	`enum-ctx-029`	50	500	recall 0.94, 3 missing, 3 wrong	2/3*
MiMo-V2.6 Pro	`enum-ctx-029`	50	500	recall 0.92, 4 missing	1/3
MiMo-V2.6 Flash	`enum-ctx-035`	100	1000	recall 0.52, 46 missing, 2 near-miss, 19 wrong, 371 made-up, 34 duplicate	2/3
DeepSeek V4 Flash	`enum-ctx-018`	5	50	recall 0.80, 1 near-miss	0/3

Table 10: Selected failures, each re-run three more times. *Reproduced*: re-runs again not exactly correct (* = a re-run with no parseable answer, not counted). MiMo cases come from an earlier OpenCode Go run; re-runs of DeepSeek and MiMo used OpenRouter.

10. Prompt variants

Asking the model to be exhaustive, or to check every record before answering, made no significant difference for any model: every interval includes zero.

Model	Qs	Exhaustive − direct (pp)	Qs	Row-by-row − direct (pp)
DeepSeek V4 Pro	–	–	–	–
DeepSeek V4 Flash	93	−2.2 [-5.4, +0.0]	93	+0.0 [+0.0, +0.0]
Claude Haiku 5.5	93	+1.1 [-3.2, +5.4]	93	−2.2 [-6.5, +2.2]
MiMo-V2.6 Pro	93	+1.1 [-5.4, +6.5]	93	+4.3 [-1.1, +9.7]
MiMo-V2.6 Flash	93	+0.0 [-8.6, +8.6]	93	+6.5 [-2.2, +15.1]

Table 11: Paired change in exact-set accuracy (percentage points) from the direct prompt (E2). DeepSeek V4 Pro's E2 runs are pending (provider usage limits).

11. Reasoning effort

The models that stay complete reason much longer on larger answer sets: from 1,050 to 8,832 tokens (median) for DeepSeek V4 Pro, and from 173 to 6,442 for Claude Haiku 5.5. MiMo-V2.6 Flash often answers after a single sentence; its answers with under 100 reasoning tokens were exact 62% of the time, against 72% for the rest. This is an association, not a controlled result.

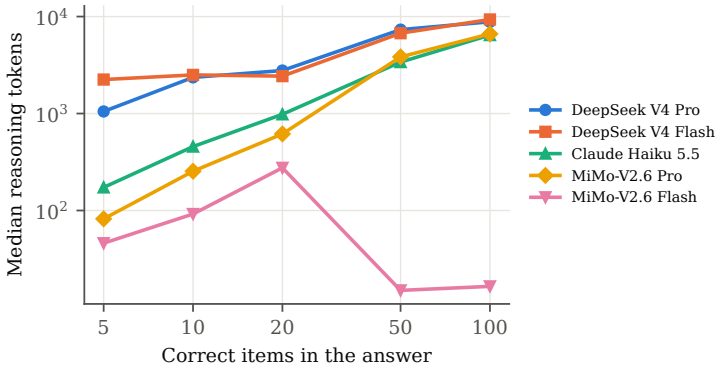


Figure 3: Median reasoning tokens by number of correct items (list-all-matches questions in context).

12. Position in the table

MiMo-V2.6 Flash misses rows near the end of a table more often than rows near the start. The other models miss too few rows to show a pattern, and no model shows a "lost in the middle" U-shape.

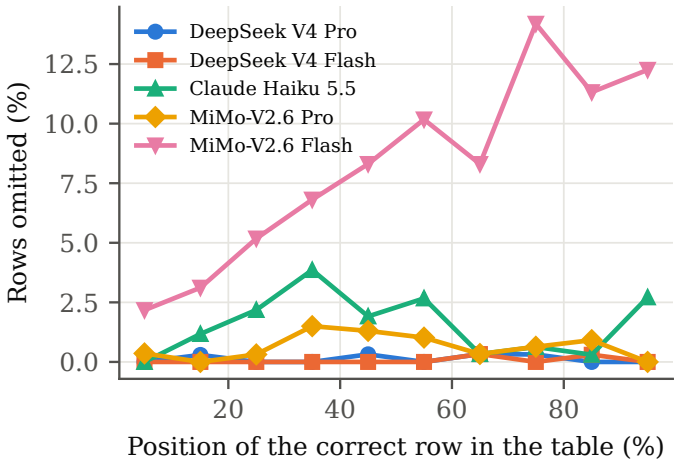


Figure 4: Share of correct rows omitted, by the row's position in the table.

13. Invalid answers and usage

7 of the 2,500 E1 answers were invalid, counted as incorrect above: 2 were refused by a safety filter, and the rest were cut off at the output limit or not parseable. Across all runs, the models answered 3,796 calls, with 1,163 further attempts failing on provider limits and being retried; 77–97% of output tokens were reasoning.

Model	Qs	Truncated	Unparseable	Refused	API error
DeepSeek V4 Pro	500	1	0	0	0
DeepSeek V4 Flash	500	1	0	0	0
Claude Haiku 5.5	500	0	0	1	0
MiMo-V2.6 Pro	500	1	0	1	0
MiMo-V2.6 Flash	500	0	2	0	0

Table 12: Invalid answers among the 500 E1 questions.

Model	Access	Calls	Failed	Input (M)	Output (M)	Reasoning (%)	Cost (USD)	Latency (s)
DeepSeek V4 Pro	OpenCode Go	594	565	3.83	2.21	97	6.87	23.4
DeepSeek V4 Flash	OpenCode Go	800	507	5.08	3.02	93	2.52	12.5
Claude Haiku 5.5	Claude Code CLI	802	91	8.04	1.88	87	2.49	7.7
MiMo-V2.6 Pro	OpenRouter	800	0	5.77	2.00	89	3.29	16.6
MiMo-V2.6 Flash	OpenRouter	800	0	5.78	1.72	77	1.25	13.1

Table 13: Usage across all runs. *Calls*: model responses, including refusals. *Failed*: provider errors, retried. Input and output in millions of tokens. Cost at list prices (Claude Haiku 5.5 and the OpenCode Go models were used through subscriptions). Latency: median per call. †Same model through a second provider.

Caveats.

- The data is synthetic and comes from a single random seed.
- E1 is one run per question; 0–13% of verdicts change on a second run.
- The answer-size trend rests on 100 questions per model.
- DeepSeek V4 Pro's E2 runs are pending (provider usage limits).

Code, data and raw responses: https://github.com/daryl-888/llm-relational-completeness.